

PA403239X

COMFORTIS™ PLUS

(spinosad + milbemycin oxime) Chewable Tablets

Product Description:
Mottled tan to brown, round, standard, biconvex tablets with letter T/number 7 debossed on one side of the tablet but no imprint on the other side.

Pharmacodynamics
The primary target of action of spinosad, a component of COMFORTIS PLUS, is an activation of nicotinic acetylcholine receptors (nAChRs) in insects. Spinosad does not interact with known insecticidal binding sites of other nicotinic or GABAergic insecticides such as neonicotinoids, fiproles, milbemycins, avermectins and cyclodienes. Insects treated with spinosad show involuntary muscle contractions and tremors resulting from activation of motor neurons. Prolonged spinosad-induced hyperexcitation results in prostration, paralysis and flea death. The selective toxicity of spinosad between insects and vertebrates may be conferred by the differential sensitivity of the insect versus vertebrate nAChRs.
Milbemycin oxime, a component of COMFORTIS PLUS, acts by binding to glutamate-gated chloride ion channels in invertebrate nerve and muscle cells. Increased permeability by the cell membrane to chloride ions causes hyperpolarization of affected cells and subsequent paralysis and death of the intended parasites. Milbemycin oxime may also act by disrupting the transmission of invertebrate neurotransmitters, notably gamma amino butyric acid (GABA).

Pharmacokinetics
Major factors of spinosad is comprised of spinosyn A and D whereas major factors of milbemycin oxime is comprised of milbemycin A₃ and A₄ 5-oximes. The mean calculated oral bioavailability was 95.8% for spinosyn A and 92.2% for spinosyn D following administration of spinosad API at 30 mg/kg in a gel capsule when co-administered with 0.5 mg/kg milbemycin oxime. For spinosyns A and D, the mean calculated volume of distribution was approximately 60 L/kg and the clearance was approximately 250 mL/hr/kg. The calculated oral bioavailability for milbemycin A₃ and A₄ 5-oximes was >100%. Specifically, the mean calculated oral bioavailability was 157% for milbemycin A₄ 5-oxime and 160% for milbemycin A₃ 5-oxime following administration of 0.5 mg/kg milbemycin oxime API in a gel capsule co-administered with spinosad at 30 mg/kg. A review of the study data could not ascribe any errors associated with the formulation preparation; dose administration; bioanalytical analysis or pharmacokinetic analysis that could have contributed to a calculated bioavailability of greater than 100%. As such, the clinical meaning or interpretation of the reported values for the absolute bioavailability for the major factors of milbemycin oxime should be done with appropriate caution. As the elimination half-lives for milbemycin A₃ and A₄ 5-oximes were comparable following IV and oral administration, it is likely that the observed increase in oral bioavailability occurs in the absorption and/or distribution phase and not in the metabolism or elimination phase of the molecule. While no definitive causality was determined, the observed high oral bioavailability (greater than 100%) could be due to the role of a transporter mechanism (i.e. P-gp) primarily at the intestine or due to mucosal metabolism of milbemycin oxime that could be inhibited by spinosad, or another unknown mechanism. The mean calculated volume of distribution was approximately 4 L/kg for milbemycin A₃ and A₄ 5-oximes and the clearance was approximately 90 mL/hr/kg and 40 mL/hr/kg for milbemycin A₃ and A₄ 5-oximes, respectively.

Indications:
COMFORTIS PLUS is indicated for the prevention of heartworm disease (*Dirofilaria immitis*). COMFORTIS PLUS kills fleas and is indicated for the prevention and treatment of flea infestations (*Ctenocephalides felis*), and the treatment and control of adult hookworm (*Ancylostoma caninum*), adult roundworm (*Toxocara canis* and *Toxascaris leonine*) and adult whipworm (*Trichuris vulpis*) infections in dogs and puppies 8 weeks of age or older and 2.3 kg of body weight or greater.

Dosage and Administration:
COMFORTIS PLUS is given orally, once a month at the minimum dosage of 13.5 mg/lb (30 mg/kg) spinosad and 0.2 mg/lb (0.5 mg/kg) milbemycin oxime body weight. For heartworm prevention, give once monthly for at least 3 months after exposure to mosquitoes (see **EFFECTIVENESS**).

Body Weight	Spinosad Per Tablet (mg)	Milbemycin oxime Per Tablet (mg)	Tablets Administered
2.3 - 4.5 kg	140	2.3	One
4.6 - 9 kg	270	4.5	One
9.1 - 18 kg	560	9.3	One
18.1 - 27 kg	810	13.5	One
27.1 - 54 kg	1620	27	One
Over 54 kg	Administer the appropriate combination of tablets		

Administer COMFORTIS PLUS with food for maximum effectiveness. To ensure heartworm prevention, owners should observe the dog for one hour after dosing. If vomiting occurs within an hour of administration, redose with another full dose. If a dose is missed and a monthly interval between doses is exceeded, then immediate administration of COMFORTIS PLUS with food and resumption of monthly dosing will minimize the opportunity for the development of adult heartworm infections and flea reinfestations.

Heartworm Prevention:
COMFORTIS PLUS should be administered at monthly intervals beginning within 1 month of the dog's first seasonal exposure and continuing until at least 3 months after the dog's last seasonal exposure to mosquitoes (see **EFFECTIVENESS**). COMFORTIS PLUS may be administered year round without interruption. When replacing another heartworm preventative product, the first dose of COMFORTIS PLUS should be given within a month of the last dose of the former medication.

Flea Treatment and Prevention:
Treatment with COMFORTIS PLUS may begin at any time of the year, preferably starting one month before fleas become active and continuing monthly through the end of flea season. In areas where fleas are common year-round, monthly treatment with COMFORTIS PLUS should continue the entire year without interruption. To minimize the likelihood of flea reinfestation, it is important to treat all animals within a household with an approved flea protection product.

Intestinal Nematode Treatment and Control:
COMFORTIS PLUS also provides treatment and control of roundworms (*T. canis*, *T. leonina*), hookworms (*A. caninum*) and whipworms (*T. vulpis*). Dogs may be exposed to and can become infected with roundworms, whipworms and hookworms throughout the year, regardless of season or climate. Clients should be advised of measures to be taken to prevent reinfection with intestinal parasites.

Contraindications:
There are no known contraindications to the use of COMFORTIS PLUS.

Warnings:
Not for human use. Keep this and all drugs out of the reach of children. Serious adverse reactions have been reported following concomitant extra-label use of ivermectin with spinosad alone, a component of COMFORTIS PLUS (see **ADVERSE REACTIONS**).

Precautions:
Treatment with fewer than 3 monthly doses after the last exposure to mosquitoes may not provide complete heartworm prevention (see **EFFECTIVENESS**). Prior to administration of COMFORTIS PLUS, dogs should be tested for existing heartworm infection. At the discretion of the veterinarian, infected dogs should be treated with an adulticide to remove adult heartworms. COMFORTIS PLUS is not effective against adult *D. immitis*. While the number of circulating microfilariae may decrease following treatment, COMFORTIS PLUS is not indicated for microfilariae clearance (see **ANIMAL SAFETY**).
Mild, transient hypersensitivity reactions manifested as labored respiration, vomiting, salivation and lethargy, have been noted in some dogs treated with milbemycin oxime carrying a high number of circulating microfilariae. These reactions are presumably caused by release of protein from dead or dying microfilariae.
Use with caution in breeding females (see **ANIMAL SAFETY**). The safe use of COMFORTIS PLUS in breeding males has not been evaluated.
Use with caution in dogs with pre-existing epilepsy (see **ADVERSE REACTIONS**).
Puppies less than 14 weeks of age may experience a higher rate of vomiting (see **ANIMAL SAFETY**).

Interaction with Other Medicaments:
COMFORTIS PLUS has been given safely with a wide variety of products and medications.

Adverse Reactions:
In a well-controlled US field study, which included a total of 352 dogs (176 treated with COMFORTIS PLUS and 176 treated with an active control), no serious adverse reactions were attributed to administration of COMFORTIS PLUS. All reactions were regarded as mild. Over the 180-day study period, all observations of potential adverse reactions were recorded. Reactions that occurred at an incidence >1% (average monthly rate) within any of the 6 months of observation are presented in the following table. The most frequently reported adverse reaction in dogs in the COMFORTIS PLUS group was vomiting.

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Chewable Tablets

Average Monthly Rate (%) of Dogs With Adverse Reactions

Adverse Reaction	COMFORTIS PLUS Chewable Tablets*	Active Control Tablets*
Vomiting	6.13	3.08
Pruritus	4.00	4.91
Lethargy	2.63	1.54
Diarrhea	2.25	1.54
Dermatitis	1.47	1.45
Skin Reddening	1.37	1.26
Decreased appetite	1.27	1.35
Pinnal Reddening	1.18	0.87

*n=176 dogs

In the US field study, one dog administered COMFORTIS PLUS experienced a single mild seizure 2½ hours after receiving the second monthly dose. The dog remained enrolled and received four additional monthly doses after the event and completed the study without further incident. Following concomitant extra-label use of ivermectin with spinosad alone, a component of COMFORTIS PLUS, some dogs have experienced the following clinical signs: trembling/twitching, salivation/drooling, seizures, ataxia, mydriasis, blindness and disorientation. Spinosad alone has been shown to be safe when administered concurrently with heartworm preventatives at label directions. In US and European field studies, no dogs experienced seizures when dosed with spinosad alone at the therapeutic dose range of 13.5 - 27.3 mg/lb (30 - 60 mg/kg), including 4 dogs with pre-existing epilepsy. Four epileptic dogs that received higher than the maximum recommended dose of 27.3 mg/lb (60 mg/kg) experienced at least one seizure within the week following the second dose of spinosad, but no seizures following the first and third doses. The cause of the seizures observed in the field studies could not be determined.

Post Approval Experience (Mar 2012):

The following adverse reactions are based on post-approval adverse drug event reporting. The adverse reactions are listed in decreasing order of frequency: vomiting, depression/lethargy, pruritus, anorexia, diarrhea, trembling/shaking, ataxia, seizures, hypersalivation, and skin reddening.

Effectiveness:

Heartworm Prevention:

In a well-controlled laboratory study, COMFORTIS PLUS was 100% effective against induced heartworm infections when administered for 3 consecutive monthly doses. Two consecutive monthly doses did not provide 100% effectiveness against heartworm infection. In another well-controlled laboratory study, a single dose of COMFORTIS PLUS was 100% effective against induced heartworm infections.

In a well-controlled six-month US field study conducted with COMFORTIS PLUS, no dogs were positive for heartworm infection as determined by heartworm antigen testing performed at the end of the study and again three months later.

Flea Treatment and Prevention:

In a well-controlled laboratory study, COMFORTIS PLUS demonstrated 100% effectiveness on the first day following treatment and 100% effectiveness on Day 30. In a well-controlled laboratory study, spinosad, a component of COMFORTIS PLUS, began to kill fleas 30 minutes after administration and demonstrated 100% effectiveness within 4 hours. Spinosad, a component of COMFORTIS PLUS, kills fleas before they can lay eggs. If a severe environmental infestation exists, fleas may persist for a period of time after dose administration due to the emergence of adult fleas from pupae already in the environment. In field studies conducted in households with existing flea infestations of varying severity, flea reductions of 98.0% to 99.8% were observed over the course of 3 monthly treatments with spinosad alone. Dogs with signs of flea allergy dermatitis showed improvement in erythema, papules, scaling, alopecia, dermatitis/pyodermitis and pruritus as a direct result of eliminating the fleas.

Treatment and Control of Intestinal Nematode Infections:

In well-controlled laboratory studies, COMFORTIS PLUS was ≥ 90% effective in removing naturally and experimentally induced adult roundworm, whipworm and hookworm infections.

Palatability:

COMFORTIS PLUS is a flavored chewable tablet. In a field study of client-owned dogs where 175 dogs were each offered COMFORTIS PLUS once a month for 6 months, dogs voluntarily consumed 54% of the doses when offered plain as if a treat, and 33% of the doses when offered in or on food. The remaining 13% of doses were administered like other tablet medications.

Animal Safety:

COMFORTIS PLUS was tested in pure and mixed breeds of healthy dogs in well-controlled clinical and laboratory studies. No dogs were withdrawn from the field studies due to treatment-related adverse reactions.

In a margin of safety study, COMFORTIS PLUS was administered orally to 8-week-old Beagle puppies at doses of 1, 3, and 5 times the upper half of the therapeutic dose band, every 28 days for 6 dosing periods.

Vomiting was seen in all groups including control animals with similar frequency. Adverse reactions seen during the course of the study were salivation, tremors, decreased activity, coughing and vocalization.

Body weights were similar between control and treated groups throughout the study. Treatment with COMFORTIS PLUS was not associated with any clinically significant hematology, clinical chemistry or gross necropsy changes. One 5X dog had minimal glomerular lipidosis observed microscopically. The clinical relevance of this finding is unknown.

Plasma spinosyn A, spinosyn D, milbemycin A₅ 5-oxime and milbemycin A₅ 5-oxime concentrations increased throughout the study. At each dosing period, plasma spinosyn A and spinosyn D concentrations were greater than proportional across the dose range 1 to 5X. Plasma milbemycin A₅ 5-oxime concentrations appeared to be dose proportional across range 1 to 5X by the end of the study. Plasma concentrations of spinosad and milbemycin oxime indicate that expected systemic exposures were achieved throughout the study.

In an avermectin-sensitive Collie dog study, COMFORTIS PLUS was administered orally at 1, 3, and 5 times the upper half of the recommended therapeutic dose band every 28 days. No signs of avermectin sensitivity were observed after administration of COMFORTIS PLUS during the study period to avermectin-sensitive Collie dogs. The adverse reactions observed in the treatment groups were vomiting and diarrhea. Body weights in all treatment groups were comparable to the control group.

Hematology and clinical chemistry parameters showed no clinically significant changes from study start to end, and all dogs were considered healthy throughout the study.

In a heartworm positive safety study, COMFORTIS PLUS was administered orally at 1, 3, and 5 times the upper half of the therapeutic dose band to Beagle dogs with adult heartworm infections and circulating microfilariae, every 28 days for 3 treatments. Vomiting was observed in one dog in the 1X group, in three dogs in the 3X group, and in one dog in the 5X group. All but one incident of vomiting was observed on the treatment day during the first treatment cycle. The vomiting was mild and self-limiting. Hypersensitivity reactions were not observed in any of the treatment groups. Microfilariae counts decreased with treatment.

In a reproductive safety study, COMFORTIS PLUS was administered orally to female dogs at 1 and 3 times the upper half of the therapeutic dose band every 28 days prior to mating, during gestation and during a six-week lactation period. Dogs with confirmed fetal heartbeats on ultrasound examination were evaluated for reproductive safety. One 3X and one 1X group female did not become pregnant. No treatment-related adverse reactions or signs of avermectin toxicosis were noted for adult females. Adult females in the 3X group lost weight during the 6-week pre-mating period, while control group females gained weight during that time. The body weights of the treated groups were comparable to the control group during gestation and post-parturition phases of the study. Gestation length, litter average body weight, litter size, stillborn pups, pup survival and the proportion of pups with malformations were comparable between treated and control dam groups. Malformations in the 1X group included a pup with cleft palate and a littermate with anophthalmia, fused single nares, misshapen palate, hydrocephalus, omphalocele and malpositioned testes; a pup with a malformation of the anterior tip of the urinary bladder and umbilical blood vessel; and a pup with patent ductus arteriosus (PDA). Malformations in the 3X group included three littermates with PDA. Malformations in the control group included a pup with a malformed sternum and a pup with PDA and a malpositioned superior vena cava. Clinical findings in pups of the treated groups were comparable to the control group except for one 1X group pup that was smaller and less coordinated than its littermates and had tremors when excited. The relationship between spinosad and milbemycin oxime treatment and the 1X and 3X dogs that did not become pregnant, the specific pup malformations and the unthrifty 1X group pup are unknown. The incidence of cleft palate is not unexpected based on the historical data collected at the breeding site.

In a margin of safety study with spinosad alone, 6-week old Beagle puppies were administered average doses of 1.5, 4.4, and 7.4 times the maximum recommended dose at 28-day intervals over a 6-month period. Vomiting was observed across all treatments, including controls, and was observed at an increased rate at elevated doses.

Vomiting most often occurred 1 hour following administration and decreased over time and stabilized when puppies reached 14 weeks of age. There is no antidote available. In the case of adverse clinical signs and overdose, treat symptomatically.

Storage Information:

Store below 25°C (Air Conditioning) and protect from light.

How Supplied:

COMFORTIS PLUS is available in five tablet sizes. Each tablet size is available in color-coded packages of 6 tablets.

- 2.3 - 4.5 kg (140 mg spinosad and 2.3 mg milbemycin oxime)
- 4.6 - 9 kg (270 mg spinosad and 4.5 mg milbemycin oxime)
- 9.1 - 18 kg (560 mg spinosad and 9.3 mg milbemycin oxime)
- 18.1 - 27 kg (810 mg spinosad and 13.5 mg milbemycin oxime)
- 27.1 - 54 kg (1620 mg spinosad and 27 mg milbemycin oxime)

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KEY

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PRODUCT INFO

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Previous Item Code	PA402488X		

ARTWORK INFO

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